

Common Mistakes to Check

Note: This checklist was made based on the common errors that were encountered during the prototyping phase of the box. After electrical assembly of each of the PCBs it is suggested that these checks be completed before powering the circuit.

SD Card Module Orientation

- Verify that the SD card module is correctly oriented as per the PCB markings.
- Ensure that the SD card slot is accessible and aligned properly.

SD Card Presence

- Make sure that the SD card is less than 32 GB.
- Confirm that an SD card is properly inserted into the SD card module.
- Verify that the system detects the SD card during functional testing.

RTC Module Orientation

- Verify that the RTC card module is correctly oriented as per the PCB markings.
- Ensure that the RTC card slot is accessible and aligned properly.

RTC Battery Installation

- Check if the RTC module has a battery installed.
- Ensure that the battery is seated correctly and provides power when the system is unplugged.
- If the battery is missing the clock needs to be reset and programmed. (refer to the RTC programming procedure)

SCL and SDA Pin Soldering

- Ensure that the SCL and SDA pins (the two pins in the middle of the Arduino Pro Mini) are properly soldered.

- Verify continuity to ensure a reliable connection and that there are no unintended connections between 2 adjacent pins..

Complete Soldering of Arduino Pro Mini

- Check visually that each and every pin on the Arduino Pro Mini has been soldered correctly. If the problem persists, use a multimeter to check continuity for the pin that is being tested.
- Look for missing, cold, or weak solder joints, and reflow if necessary.

MOSFET Orientation

- Verify that all MOSFETs are correctly placed as per the PCB markings.
- Ensure that the drain, source, and gate pins are oriented properly to avoid incorrect functionality.

Loose or Misaligned Components

- Check for any misplaced or misaligned connectors, modules, or components before final assembly.

Unconnected or Faulty Wiring

- Ensure all wiring is firmly connected, especially power and sensor cables (data cable for the sound sensor). Keep the power cable taped to the top face of the box.
- Inspect for frayed or damaged wires that may cause short circuits or poor connectivity.

Check for the Cellular Module Antenna

- Verify that the cellular module antenna wire is properly taped to the side of the box.
- Ensure the antenna is securely mounted on its PCB connector and positioned to avoid interference with the Box door.

Operator Name:

Date:

Mechanical checks for Sensor Box:

Solar Panel Mount Assembly

- Check for 4 screws installed to secure the solar panel mount.
- Check for 4 spring washers installed behind the solar panel mount screws.
- Check for 4 nuts installed to secure the solar panel mount.
- Check for 3 metal rubber washers installed on top of the solar panel bracket mount.
- Check for 3 screws installed on the metal rubber washers on the solar panel bracket mount.
- Check for 3 plastic retainer screws for fastening the solar panel onto the mount.

Cross Member Assembly

- Check for 4 spacers installed behind the back screws for the cross members.
- Check for Loctite applied (look for the red residue on all screws) and 4 Allen bolts installed on the cross members.
- Check for Loctite applied and 4 additional screws installed for the cross members.

Solar Shield Assembly

- Check for 7 disks present for the solar shield.
- Check that all disks are securely connected using the notch guide pins between each disk.
- Check for 5-inch long screw inserted through the center of the stacked disks.

- Check for screw aligned and passed through the mounting holes in the box.
- Check for O-ring installed onto the screw from inside the box.
- Check for nut installed from inside the box to secure the assembly.

Antenna and Sealants

- Check for cellular module antenna taped securely.
- Check for silicone applied on the outer surface of the box and between the glands (look for white residue to confirm this).
- Check for glue applied to secure the mesh inside the box.
- Check for silicone applied around the solar panel charging wire entry point.

PCB Board Mounting

- Check for 6 isolating washers installed on and below the PCB standoffs.
- Check for 3 screws installed into the PCB standoffs below the mesh.
- Check for 3 screws installed into the PCB standoffs above the mesh.

Sound Sensor Mounting

- Check for 2 screws installed to secure the sound sensor.
- Check for 2 nuts installed behind the sound sensor.
- Check for 4 isolating washers installed for the sound sensor.
- Check for 2 screws installed on the mesh plate for the sound sensor mount.
- Check for 2 nuts installed above the mesh plate for the sound sensor.
- Check that all nuts and bolts are tightened properly to securely hold the sound sensor in place, back checking it using your finger and confirming that no bolts are loose.

Temperature Sensor Mounting

- Check for temperature sensor installed into the gland.
- Check for nut turned on the exterior to secure the sensor firmly.
Note: This step should be done before assembling the solar shield.

PV Panel testing

- Connect PV panel to the voltaic LED and see if it lights up.

Cable Management

- Check for 1 zip tie installed for temperature sensor cable management.
- Check for 1 zip tie installed for sound sensor and solar panel USB cable management.
- Check for the elbow connector to be connected to the USB C connector cable
- Check for the USB C connector cable for charging the power bank

Box Label Installation

- Install the QR code label on the top face of the box.
- Install the box number label directly below the QR code.
- Apply UV-resistant tape over the box label to protect them from peeling.

Box Sticker Checks

- Check for a yellow sticker on the sound sensor
- Check for a red sticker on the ADC
- Check for a green sticker on the cellular module
- Check for a number on the yellow label of the temperature sensor
- Check for a check for a yellow sticker on the SD module
- Check for a green sticker on the RTC module
- Check for a blue sticker on the PCB board

Final Quality Checks

- Ensure all screws are secured to standoffs, cables are properly connected, and there is no exposed insulation or loose hardware. If any rattling or noise is detected, indicating loose components, restart the checklist and tighten all identified loose screws, nuts, or bolts.
- Verify that the box stickers QR code and BOX number are properly affixed and visible on the exterior.
- Verify that the BOX number sticker has UV resistant tape on it.
- Check for a packet of desiccant in the BOX.

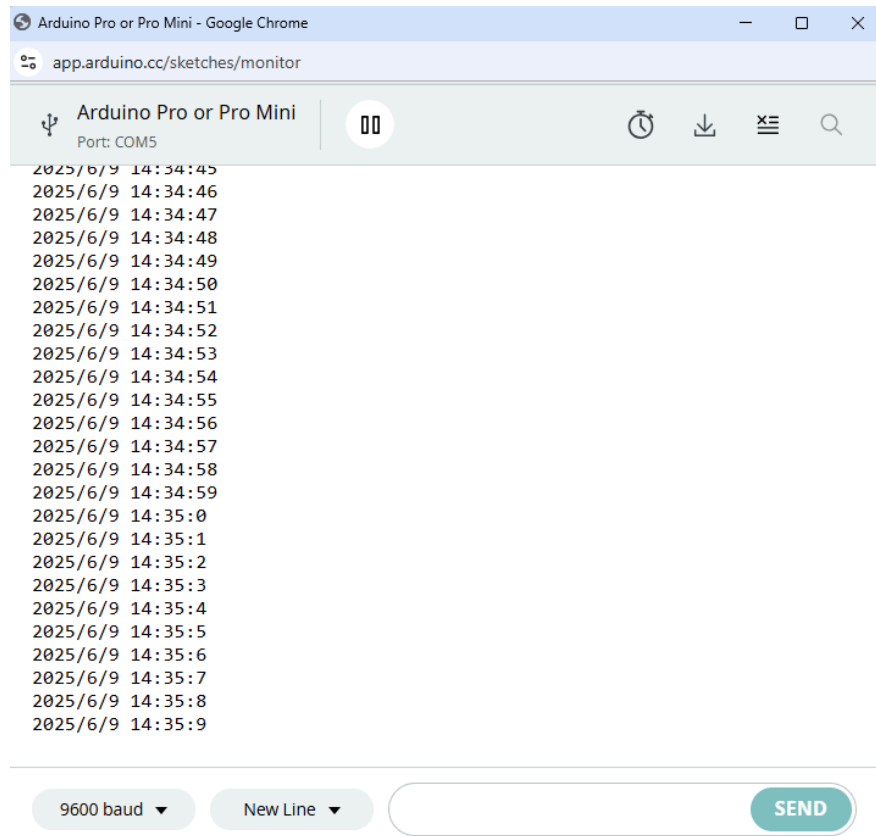
Electrical Checks for Sensor Box:

1. Module Mounting and Connections

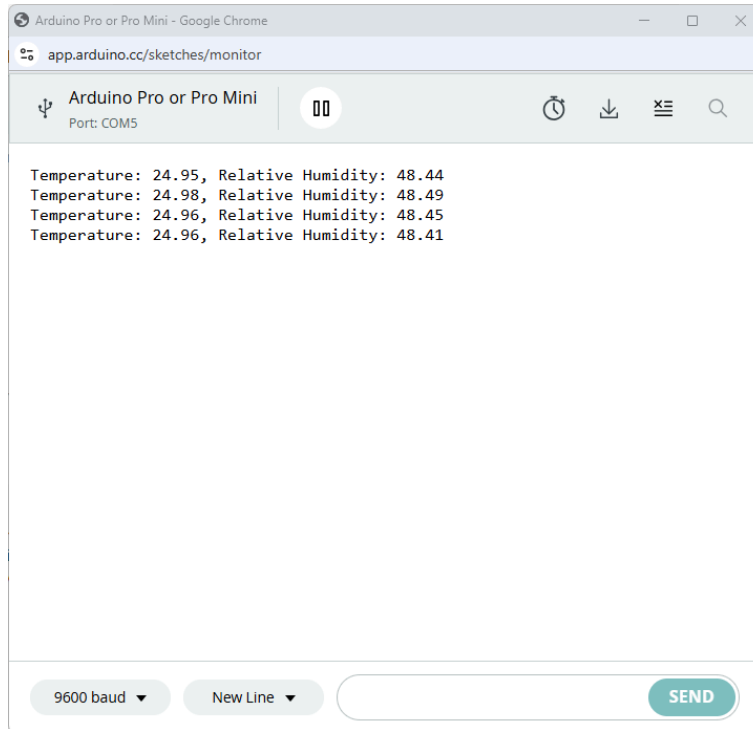
- Ensure the LTE module, Arduino, RTC, ADC and SD module are securely mounted on the PCB and in the right orientation.
- Verify that the power cable, sound sensor cable, and temperature and humidity sensor cable are firmly connected to screw terminals; no extra wire is seen outside.

2. Functional Testing Using Test Scripts

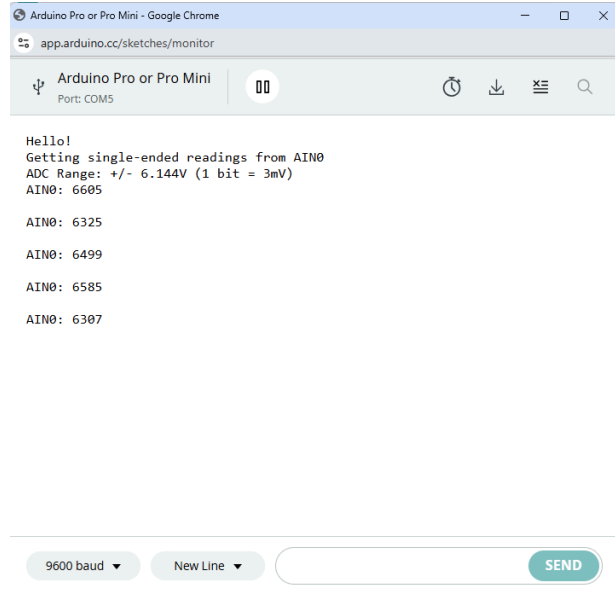
- Verify RTC
 - Upload RTC test code to Arduino
 - Open Serial Monitor of the Arduino IDE or Web Editor
 - If the Serial Monitor pops out the correct time stamps as below, that means the RTC module works fine and got proper initialized.



- Verify T&RH sensor
 - Upload T&RH test code to Arduino
 - Open Serial Monitor of the Arduino IDE or Web Editor and wait for 5 seconds initialization
 - If the Serial Monitor pops out changing readings every 2 seconds and the T and RH reading agrees with the weather station, that means the T&RH sensor works fine.



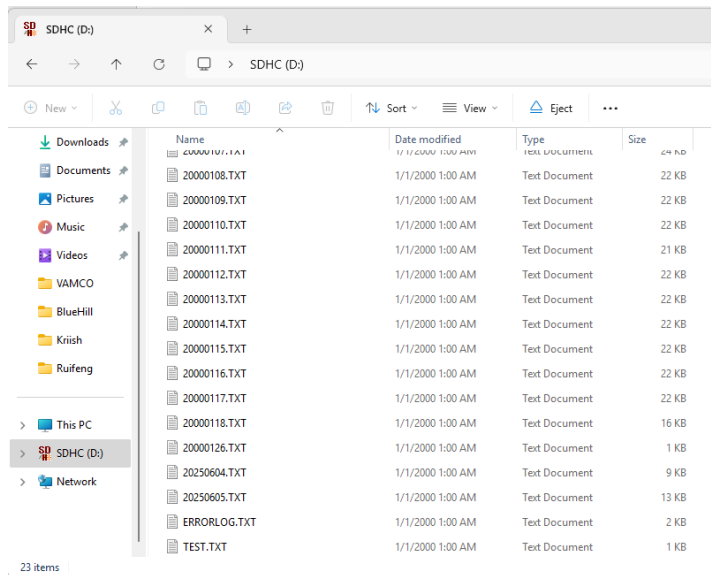
- Verify noise sensor + ADC module
 - Upload noise sensor + ADC test code to Arduino
 - Open Serial Monitor of the Arduino IDE or Web Editor
 - If the Serial Monitor pops out the reasonable noise reading as below, that means the noise sensor and the ADC module work fine.

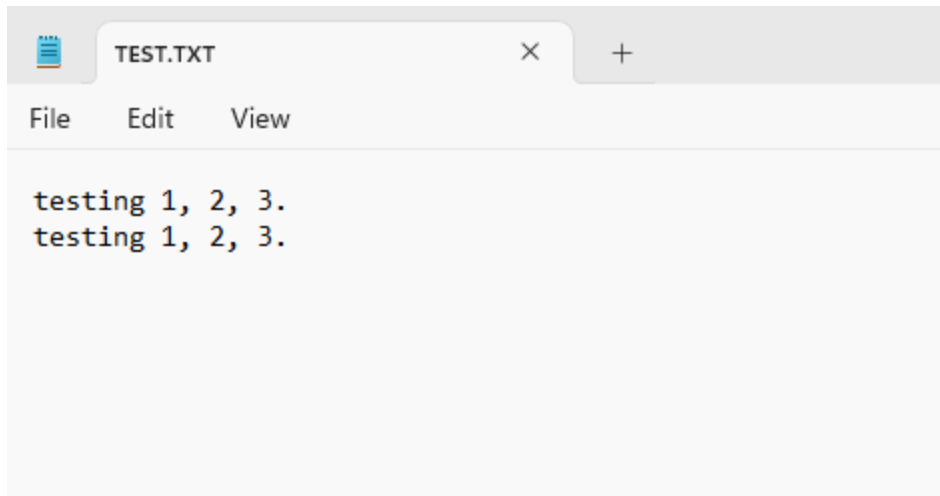


- Verify SD module
 - Upload SD module test code to Arduino
 - Open Serial Monitor of the Arduino IDE or Web Editor
 - If the Serial Monitor says, "Writing to test.txt...done.", and the "text.txt" file in the SD card contain the test message, that means the SD module works fine.



Initializing SD card...initialization done.
Writing to test.txt...done.





- Verify communication between Arduino and LTE module
 - Upload communication between Arduino and LTE module test code to Arduino
 - Open the particle.io console and click in the right LTE module number, No.19 for this example
 - If we see the left bottom area of the console pops out “Test” every 2 seconds, that means Arduino and LTE module communicate well.

A screenshot of the Particle.io console interface. The top bar shows 'Sandbox' and 'Envsensors Lab'. The main area is divided into sections: 'General information' (Name: 19, ID: e00fce6897349dcaadb7b012, Owner: envsensorslab@gmail.com, ICCID: 89883070000036202099, IMEI: 351457830679285, Serial number: P044AD30952KYB), 'State' (Status: Online, Last heard: Jun 9th 2025, 3:56 pm, Last handshake: Jun 9th 2025, 3:56 pm, Device OS: 4.0.2, Firmware: none, Groups: blue_hill), and 'Location' (You must opt in to location storage to see the location of this device). Below these is a table of events with columns: NAME, DATA, DEVICE, and PUBLISHED AT. The table shows several events, including 'Environmentdata' and 'particle/device/updates/pending'. On the right, there is a 'Last vitals' section showing cellular signal strength (45%), cellular signal quality (75%), round-trip time (709ms), battery charge (100%), cloud disconnect events (0), and rate-limited publishes (0).

NAME	DATA	DEVICE	PUBLISHED AT
Environmentdata	Test	19	6/9/25 at 3:56:29 pm
Environmentdata	Test	19	6/9/25 at 3:56:24 pm
Environmentdata	Test	19	6/9/25 at 3:56:19 pm
spark/device/diagnostics/update	[{"device":{"network":{"cellular":{"radio_a...}}	19	6/9/25 at 3:56:14 pm
Environmentdata	Test	19	6/9/25 at 3:56:14 pm
particle/device/updates/pending	false	19	6/9/25 at 3:56:13 pm
Setup	Setup Done	19	6/9/25 at 3:56:12 pm
particle/device/updates/forced	false	19	6/9/25 at 3:56:12 pm
particle/device/updates/enabled	true	19	6/9/25 at 3:56:12 pm
particle/device/updates/enabled	true	19	6/9/25 at 3:56:12 pm

5. Final Verification and Stability Testing

- Take boxes out to a place where there is good signal for the LTE module.
- Start the box by opening the box and connecting the usb cable to the powerbank.
- Ensure all data is uploaded to the cloud correctly for a continuous period of at least 60 minutes, which is 4 cycles of cloud data upload.
- If all tests have been passed attach a sticker inside the box on the right side with a date on it and include your initials. Move the box into inventory for deployment.